

REPORT – LAB 2

Course: Software Engineering

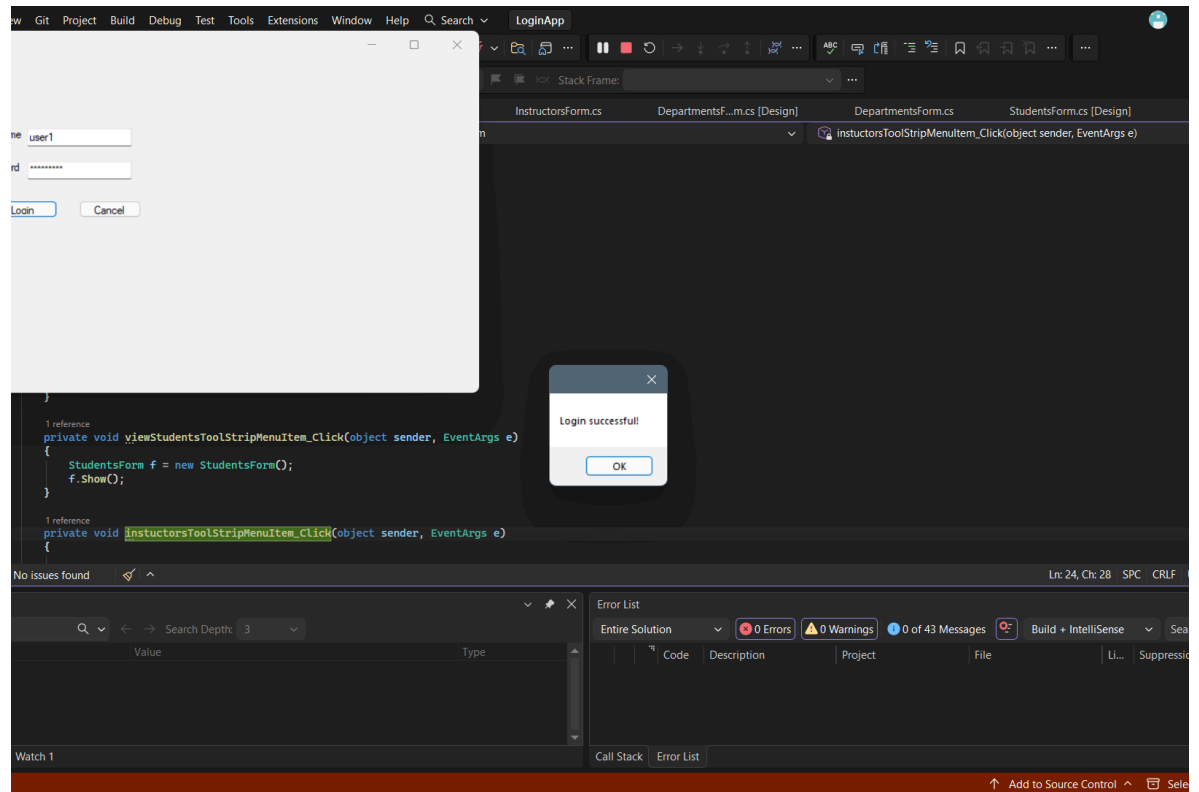
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1. Summary

- ✓ In this lab, I learned how to build a Windows Forms application and connect it to SQL Server. The goal is to create a simple management system with login functionality and CRUD operations for different tables such as Students, Courses, Instructors, and Departments.
- ✓ Through this lab, I understand how a basic database application works in practice.
- ✓ First, I created a database named Lab2DB using SQL Server and created tables such as tblUsers, tblStudents, tblCourses, tblInstructors, and tblDepartments.
- ✓ Then, I built a Login Form to check username and password from the database.
- ✓ After that, I created a Main Form with a MenuStrip to navigate between different forms.
- ✓ Next, I implemented separate forms for Students, Courses, Instructors, and Departments. Each form supports CRUD operations (Add, Refresh, Update, Delete) using DataGridView and TextBox.
- ✓ During this process, I encountered some difficulties. At first, I had trouble connecting to SQL Server due to incorrect connection strings. I also faced issues when buttons and menu items did not respond because I forgot to assign events in the Design view. In addition, I was confused when working with multiple forms and linking them through the MainForm menu.
- ✓ I also learned how to connect to the database using connection string in App.config and use SqlConnection, SqlCommand, and SqlDataAdapter to manipulate data.

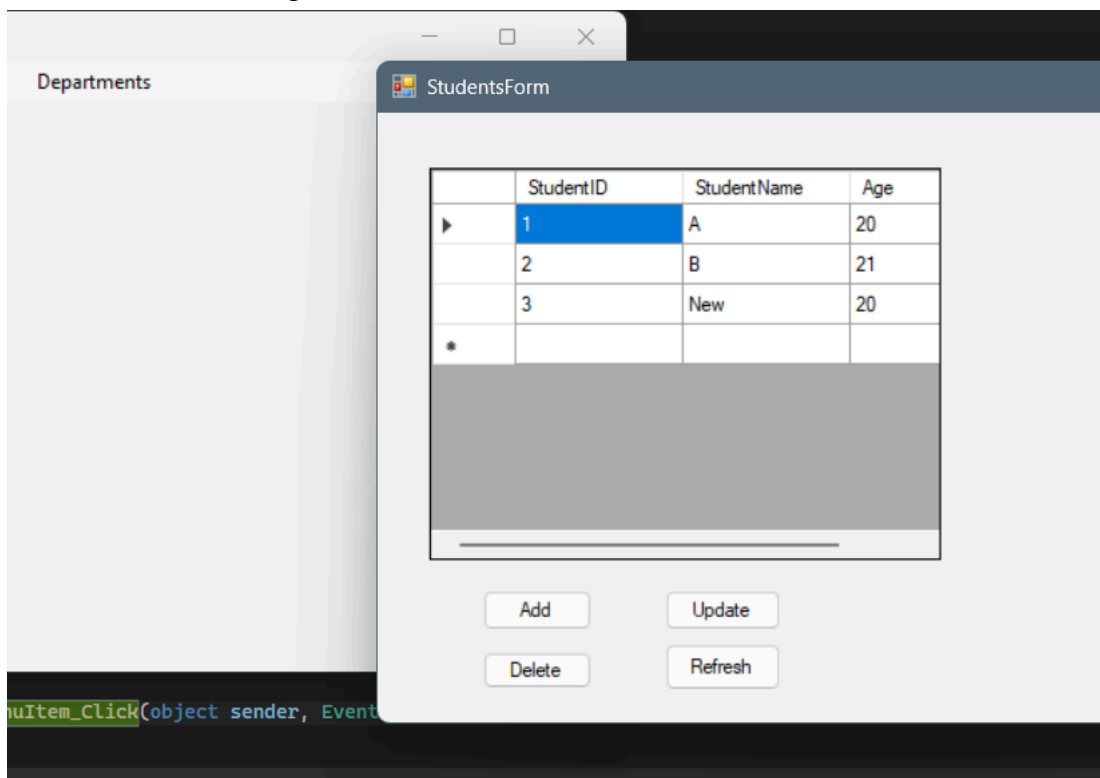
2. Homework

- ✓ Login successful :



✓

✓ Students management form:



✓ Courses management form:

The screenshot shows a Windows application with a 'LoginApp.MainForm' window and a 'CoursesForm' dialog box. The 'CoursesForm' dialog box contains a table with three columns: 'CourseID', 'CourseName', and 'Credits'. The table has three rows: the first row has '1' in 'CourseID' and 'Math' in 'CourseName'; the second row has '2' in 'CourseID' and 'Physics' in 'CourseName'; the third row is empty. Below the table, there are two text input fields and four buttons: 'Add', 'Update', 'Delete', and 'Refresh'. The 'Add' and 'Update' buttons are positioned above the 'Delete' and 'Refresh' buttons. The 'Add' button is to the right of the first text input field, and the 'Update' button is to the right of the second text input field. The 'Delete' button is to the right of the first text input field, and the 'Refresh' button is to the right of the second text input field. The 'CoursesForm' dialog box is titled 'CoursesForm'.

CourseID	CourseName	Credits
1	Math	3
2	Physics	4

Input fields and buttons: Add, Update, Delete, Refresh.

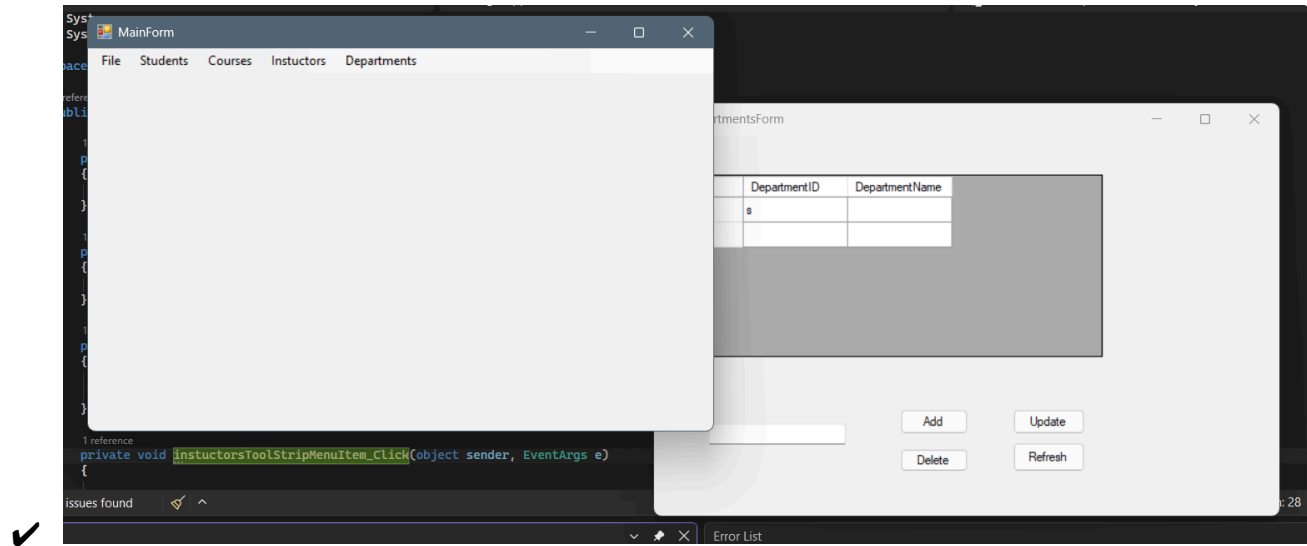
✓ Instructors management form:

The screenshot shows a Windows application with a 'LoginApp.MainForm' window and an 'InstructorsForm' dialog box. The 'InstructorsForm' dialog box contains a table with two columns: 'InstructorID' and 'InstructorName'. The table has one row with a blue header and one empty row. Below the table, there is a text input field and four buttons: 'Add', 'Update', 'Delete', and 'Refresh'. The 'Add' and 'Update' buttons are positioned above the 'Delete' and 'Refresh' buttons. The 'Add' button is to the right of the text input field, and the 'Update' button is to the right of the text input field. The 'Delete' button is to the right of the text input field, and the 'Refresh' button is to the right of the text input field. The 'InstructorsForm' dialog box is titled 'InstructorsForm'.

InstructorID	InstructorName

Input field and buttons: Add, Update, Delete, Refresh.

✓ Departments management form:



- After completing this lab, I have a better understanding of how to build a Windows Forms application connected to a database.
- I learned how to implement login functionality, perform CRUD operations, and manage multiple forms in a project.
- This lab helped me gain practical experience and improved my problem-solving skills when dealing with errors.